COMS30020 - Computer Graphics

Week 4 Briefing

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Exciting News !

And now, the moment you've all been waiting for...

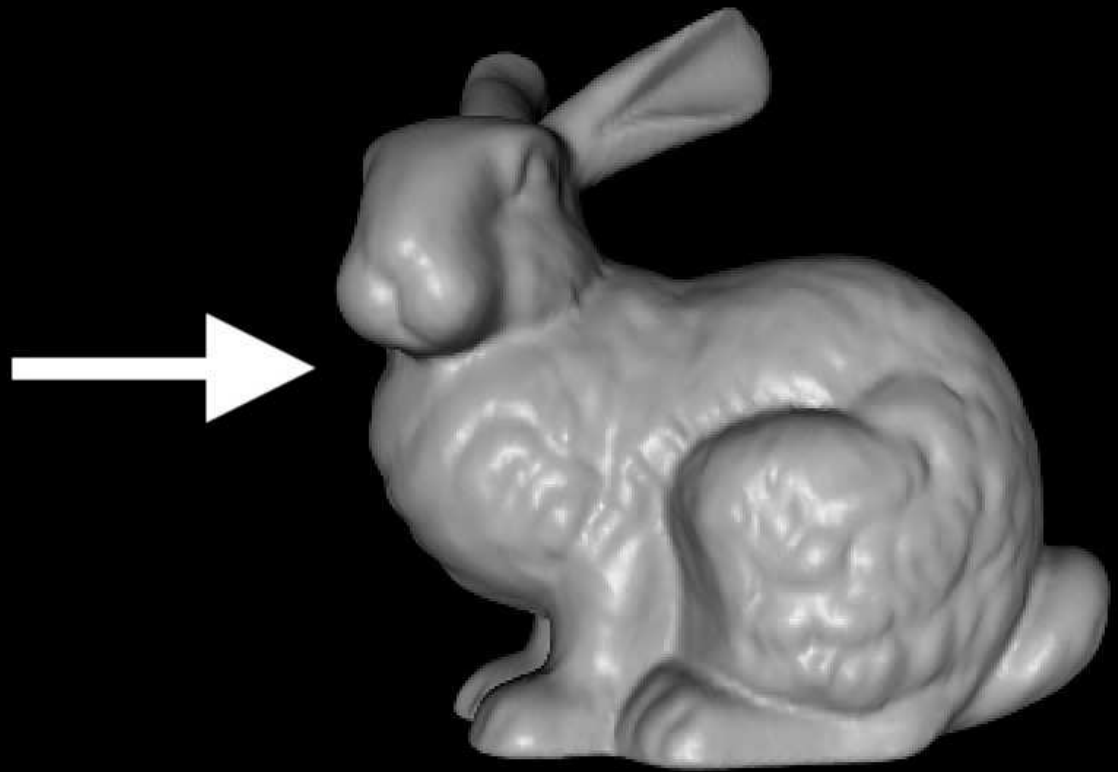
TIME TO START WORKING IN 3D !!!

2D functions you've already written are still useful !

Your Objective is Simple (to Explain ;o)

stanford-bunny.obj

```
v -0.037830 0.127940 0.004475
v -0.044779 0.128887 0.001905
v -0.068010 0.151244 0.037195
v -0.002287 0.130150 0.023220
v -0.022605 0.126675 0.007156
v -0.025108 0.125921 0.006242
v -0.037121 0.127449 0.001796
v 0.033213 0.112692 0.027686
v 0.038043 0.109755 0.016169
v -0.025508 0.112568 0.036677
v -0.024531 0.112636 0.037347
v 0.027403 0.121560 0.021221
v -0.062896 0.158419 -0.017587
v 0.040081 0.104202 0.022168
v 0.045153 0.093197 0.011160
v -0.032497 0.174231 -0.002390
```



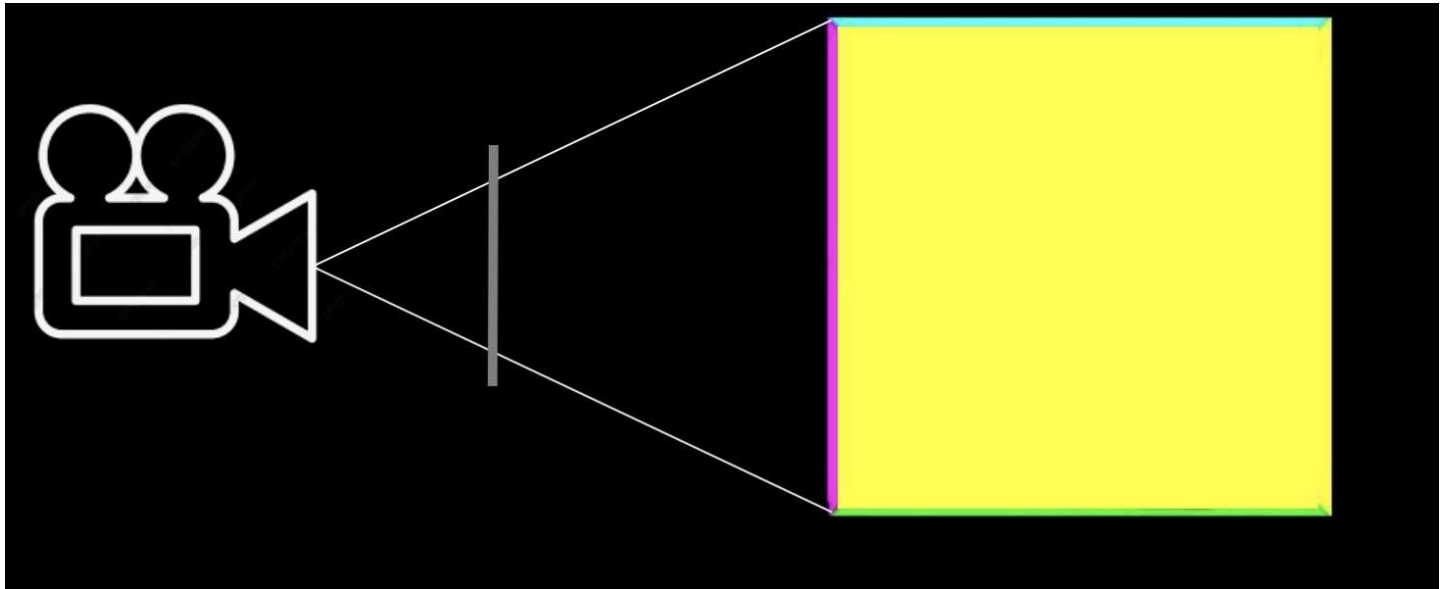
General setup for the rest of the unit

'model' a scene in three dimensional space (.OBJ)

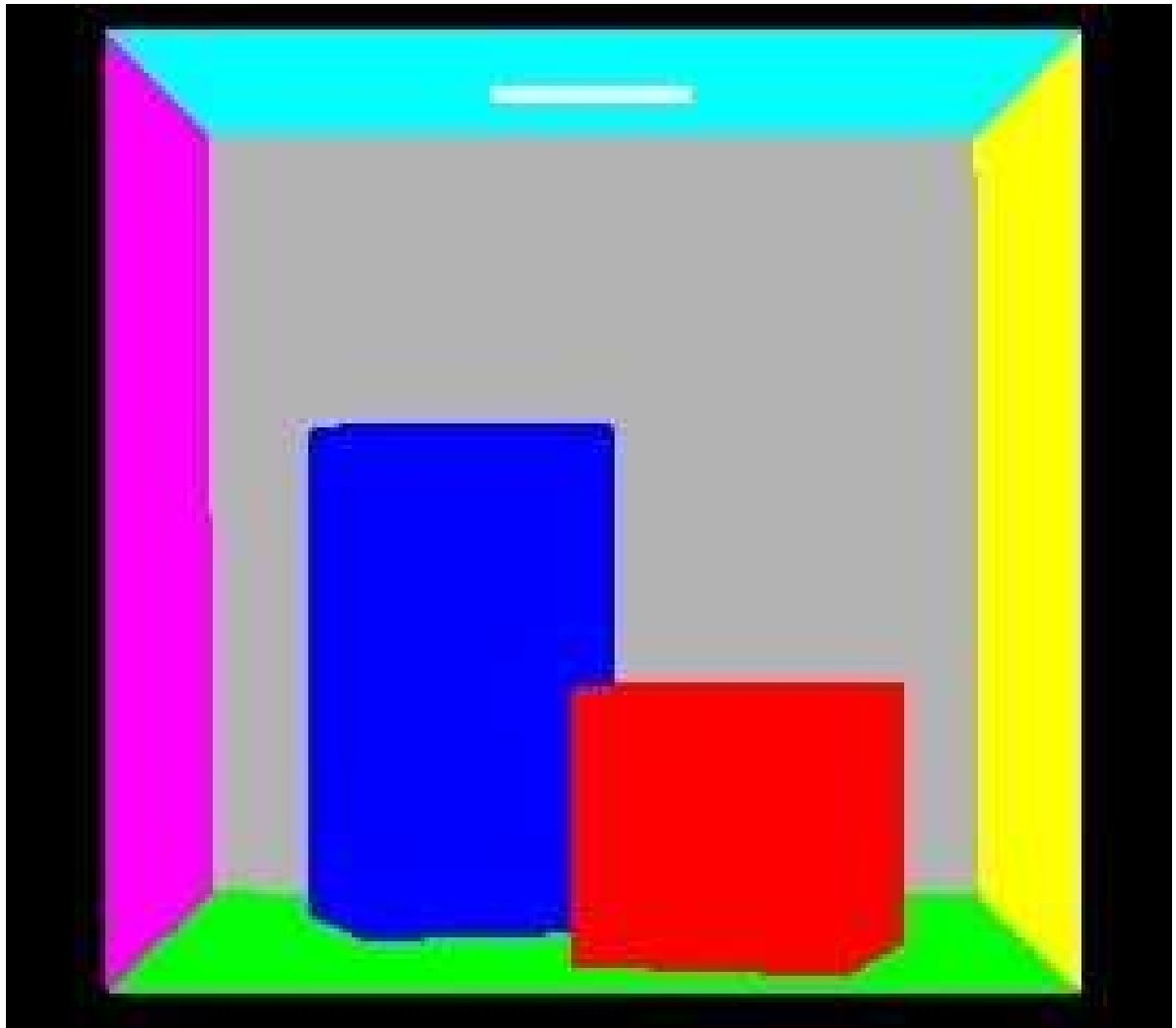
'camera' our chosen viewpoint on the model/scene

'image plane' canvas onto which you'll draw scene

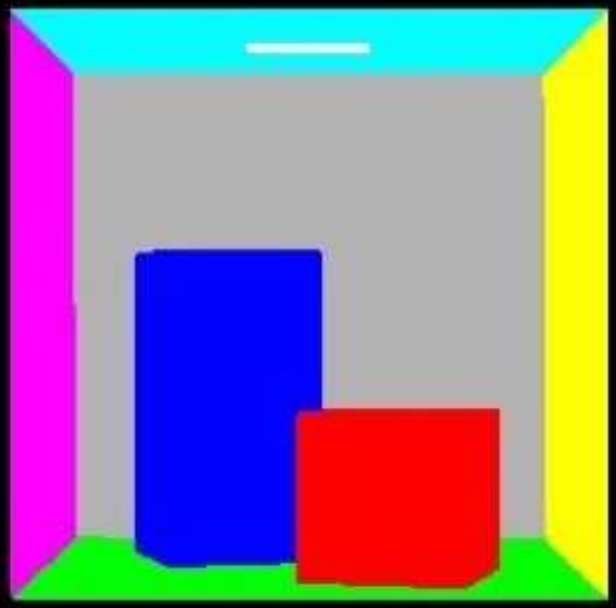
'focal length' distance from camera to image plane



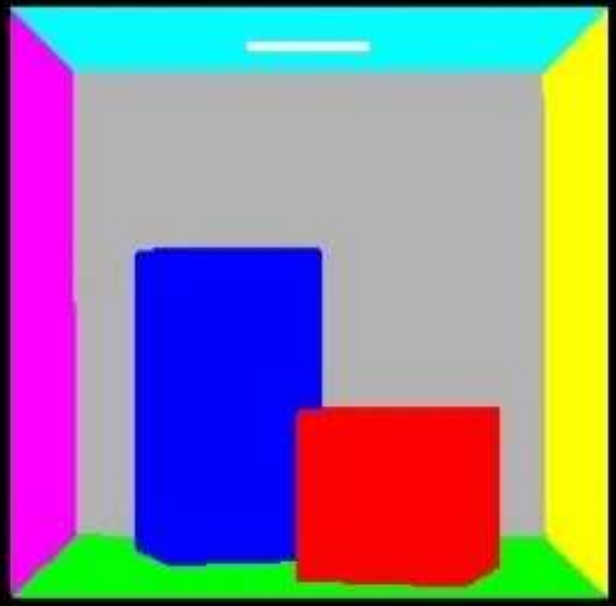
Cornell Box: Your model for next 4 weeks





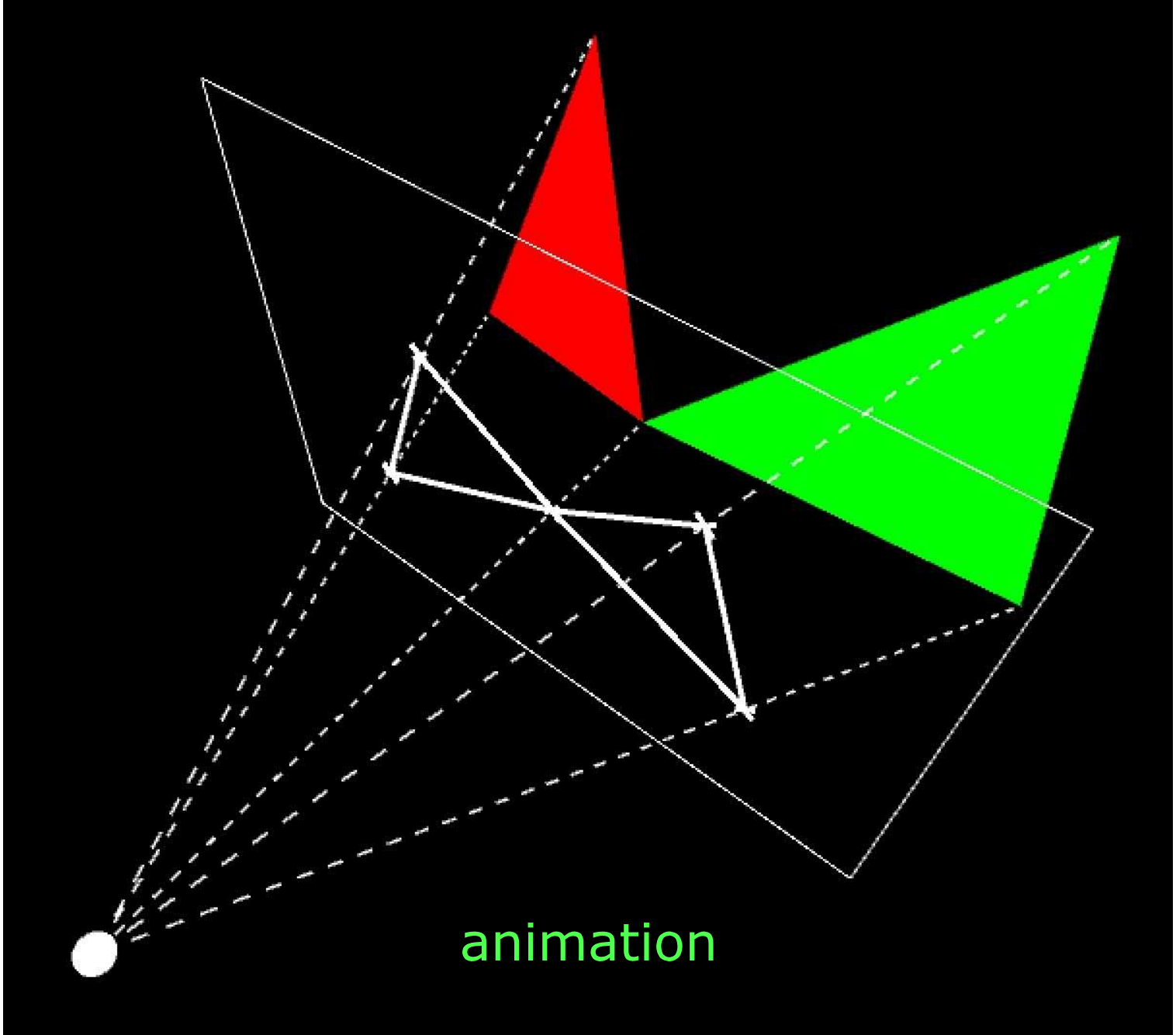




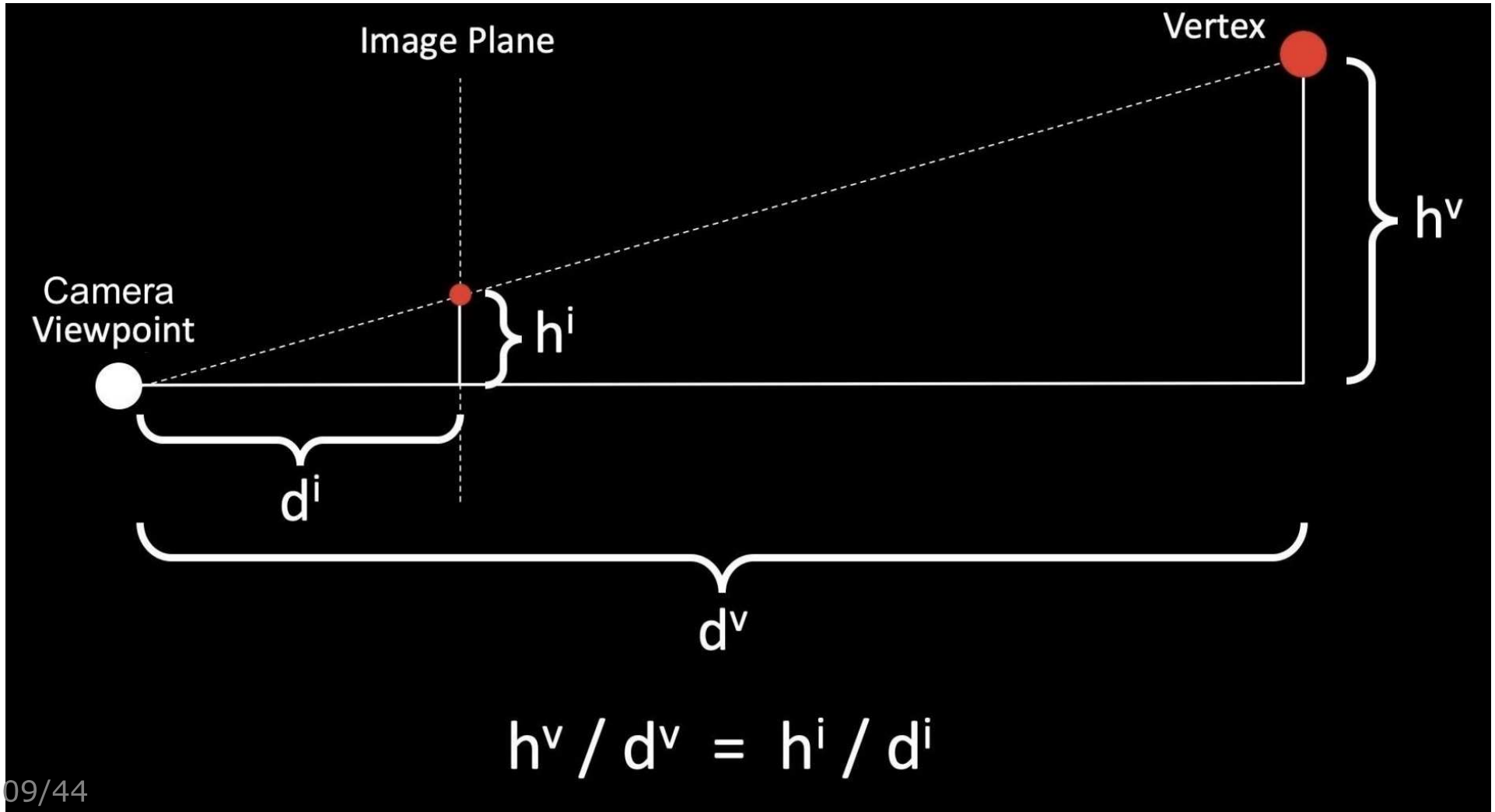


In order to render this you will need to...

- Load in some geometry data (points in 3D space)
- Load in some colour information ('materials')
- Take a "point of view" on the model (camera pos)
- Set the position of 'image plane' ('focal length')
- Do "a bit of maths" (to 'project' vertices into 2D)
- Draw model triangles onto canvas ('image plane')


animation

Approach is based on Similar Triangles

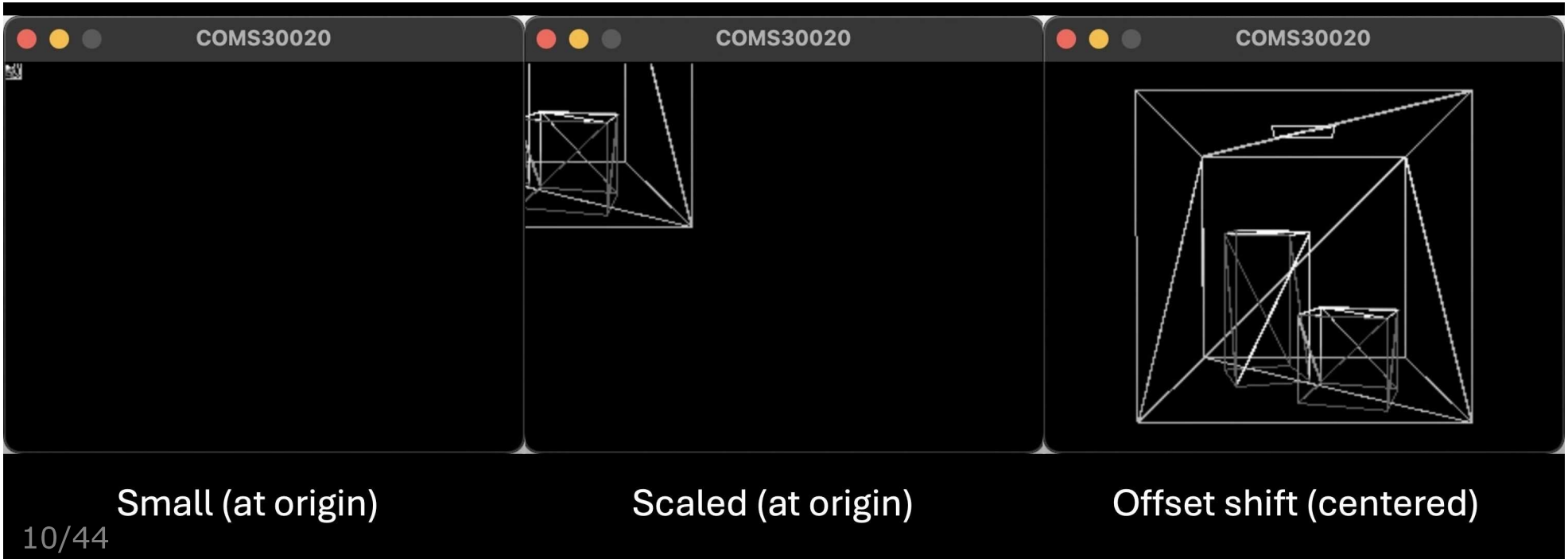


Top Tip: Break problem down into steps

Do NOT try to implement the formulae "all in one go"

Work step-by-step, using MULTIPLE lines of code

Get each step working, BEFORE moving on to the next



Identical Rendering ?

From this point on, achieving results *identical* to samples shown in workbooks becomes much harder

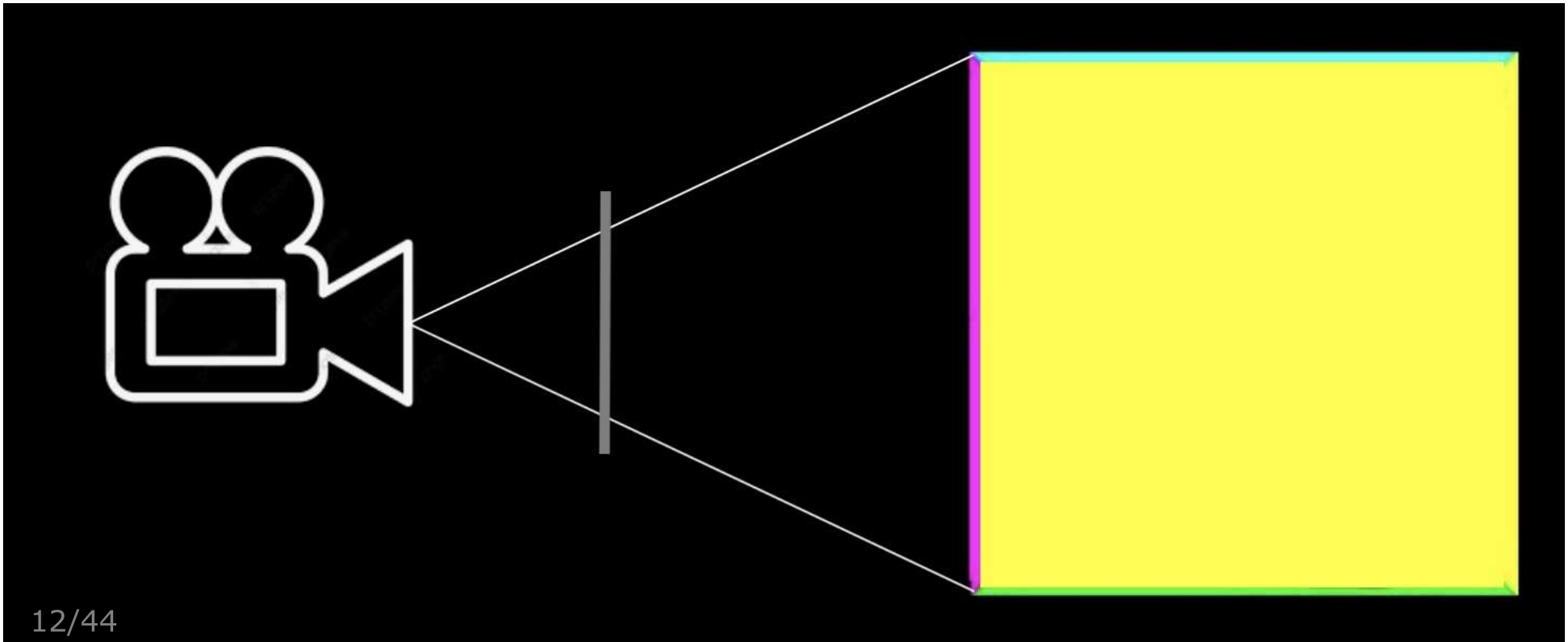
This is because the render you produce of a scene Will depend on parameters used by your renderer:

- Model scaling (when you load-in vertices)
- Model position (relative to world origin)
- Camera position (relative to world origin)
- Image plane position ('focal length')
- Image plane scaling (used to fill the window)

Let's take a look at some variations...

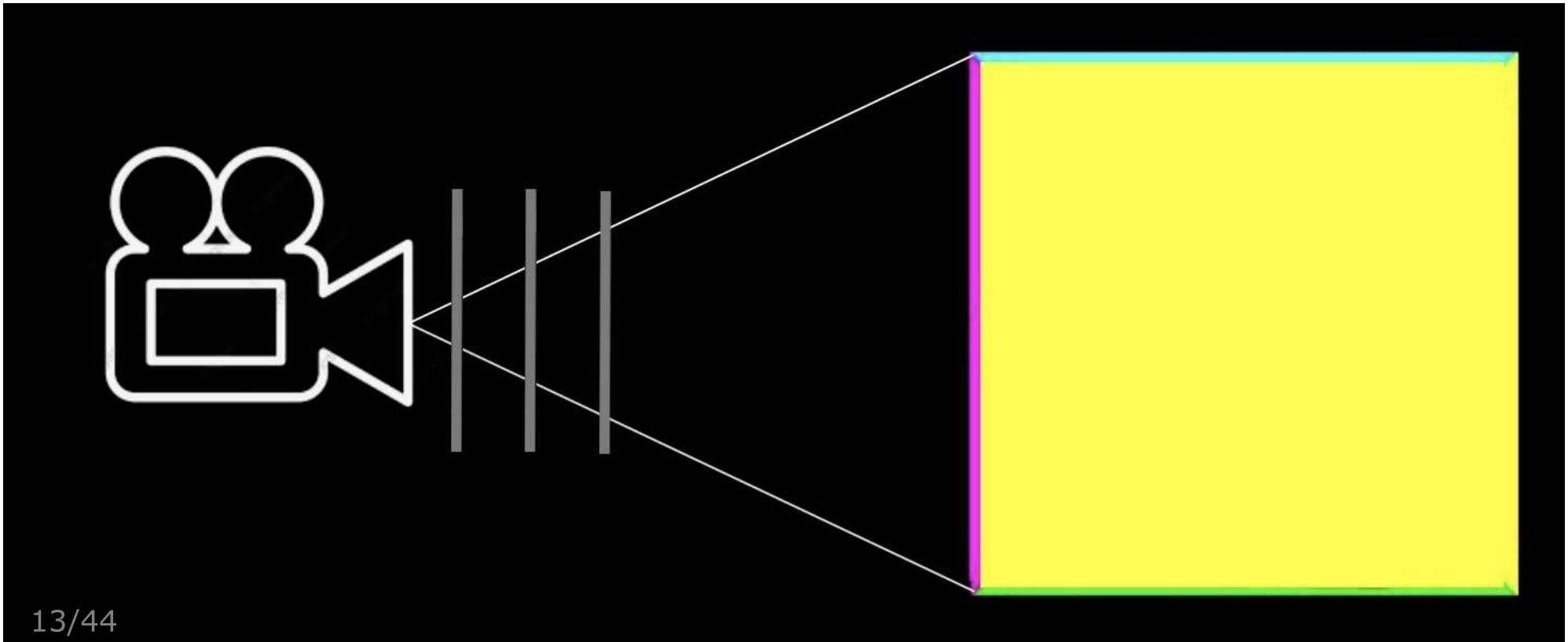
"Sight lines" help us imagine how a render will appear

What if we move just the position of the image plane ?

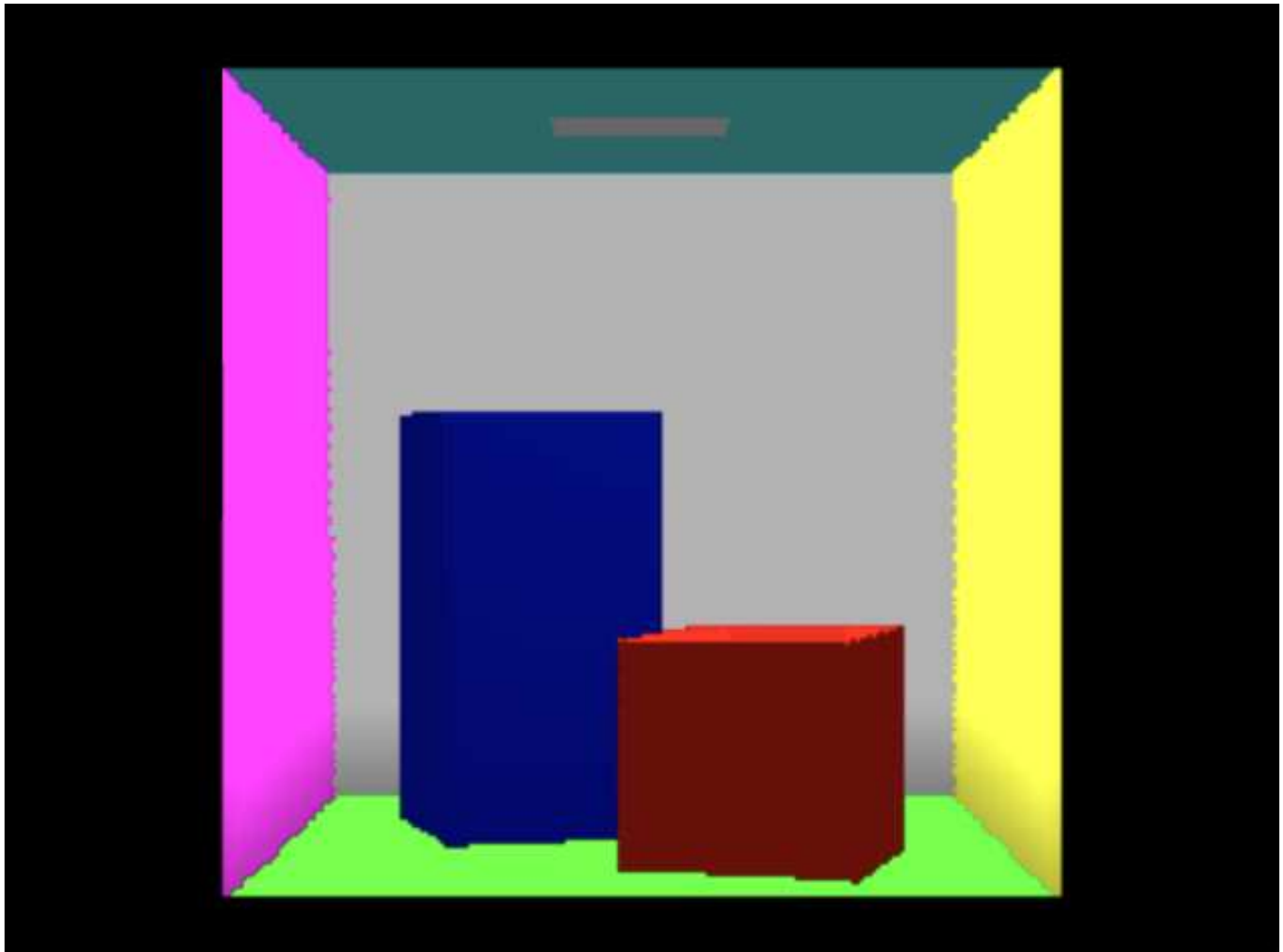


Our perspective (sight lines) on the scene stays same

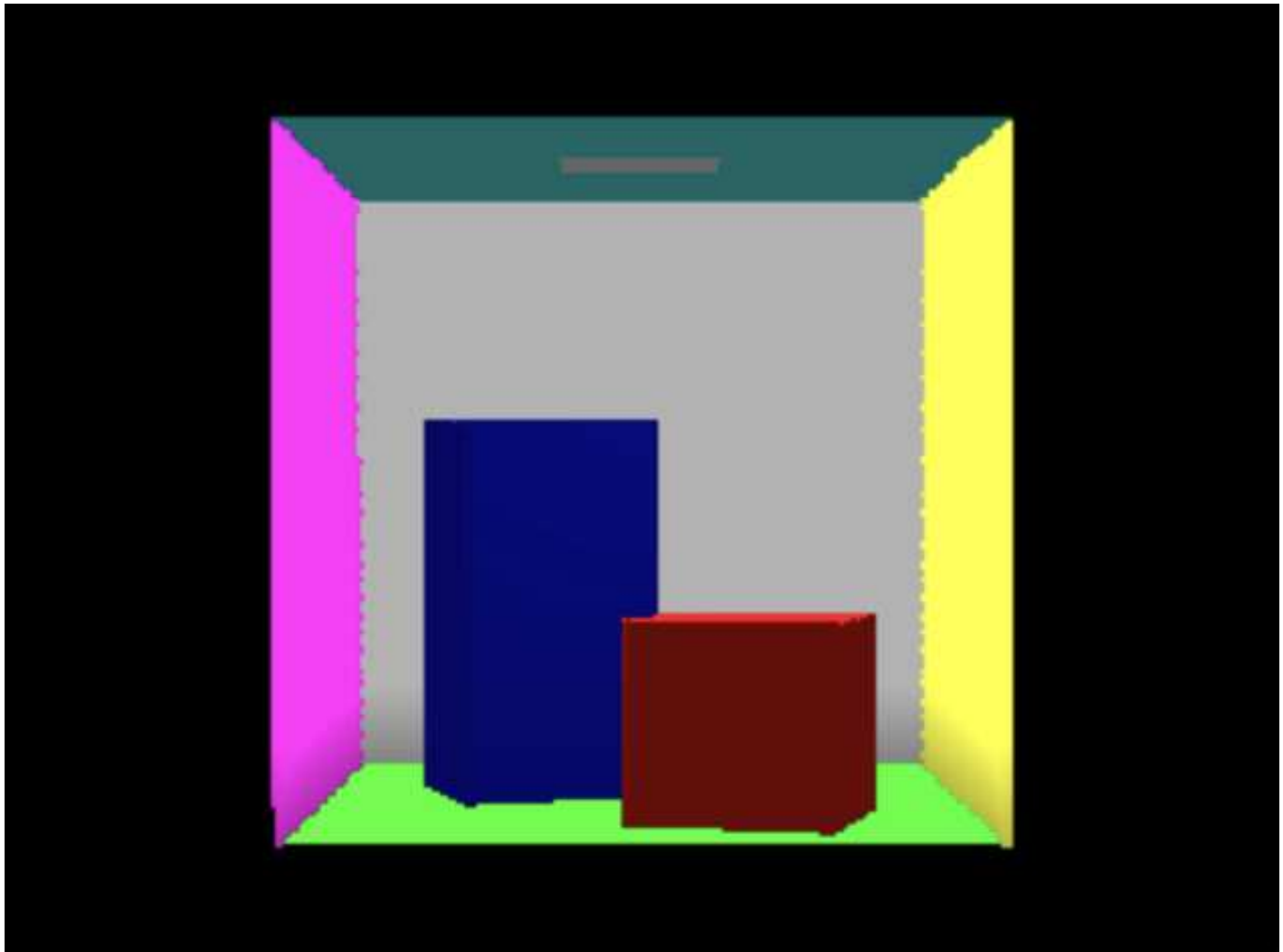
However, the size of image on image plane will change



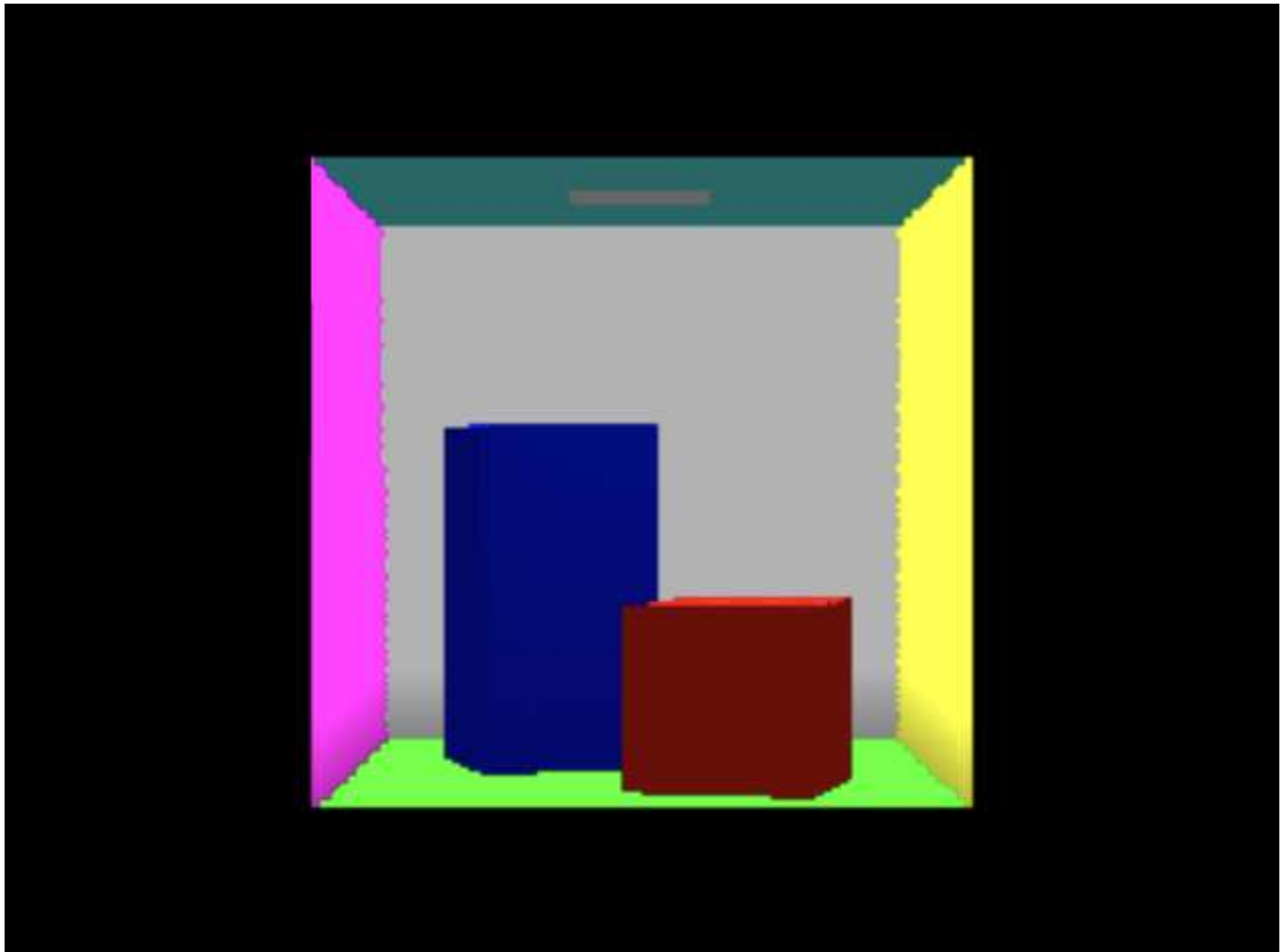
Focal Length of 4



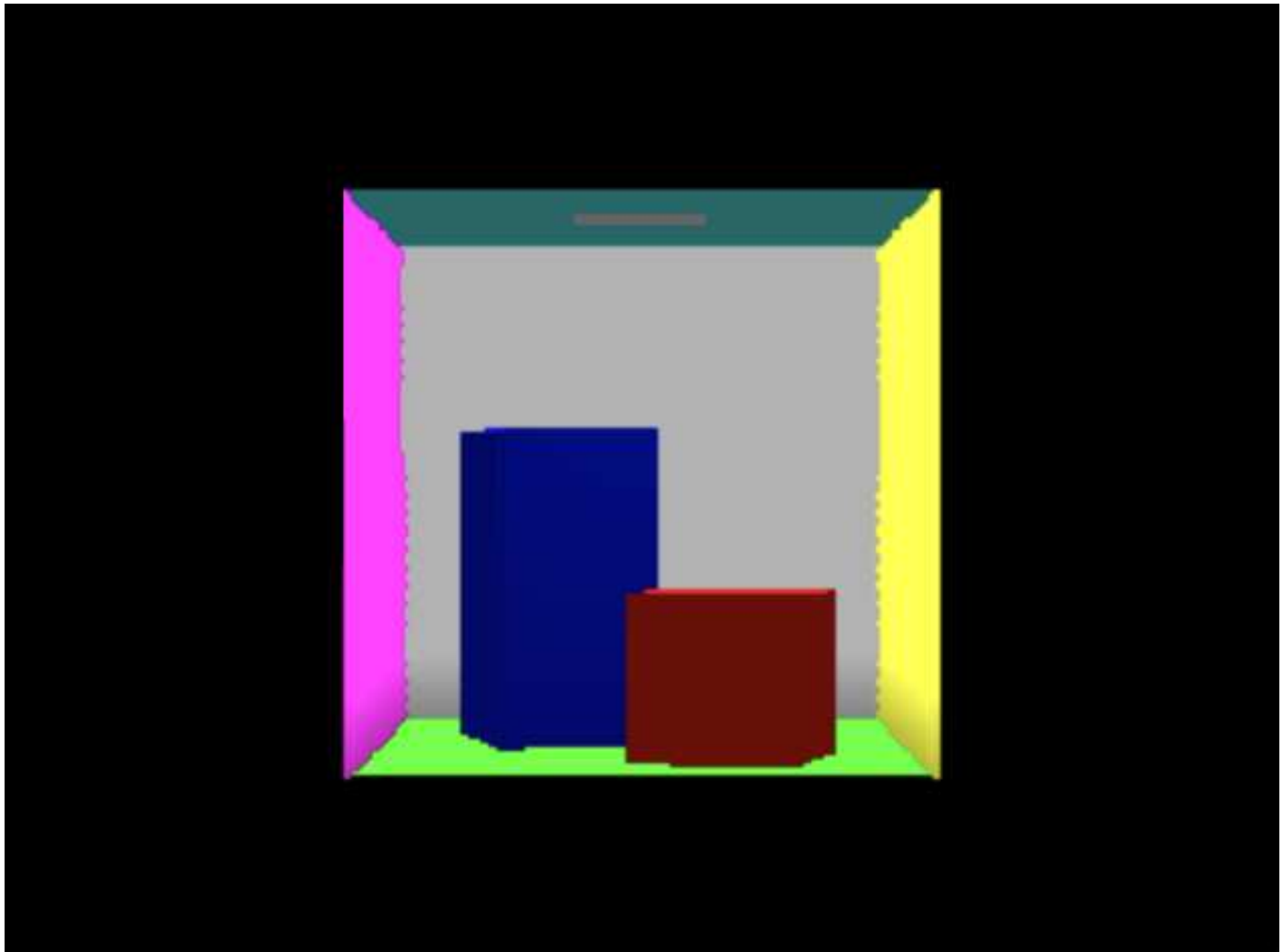
Focal Length of 3



Focal Length of 2



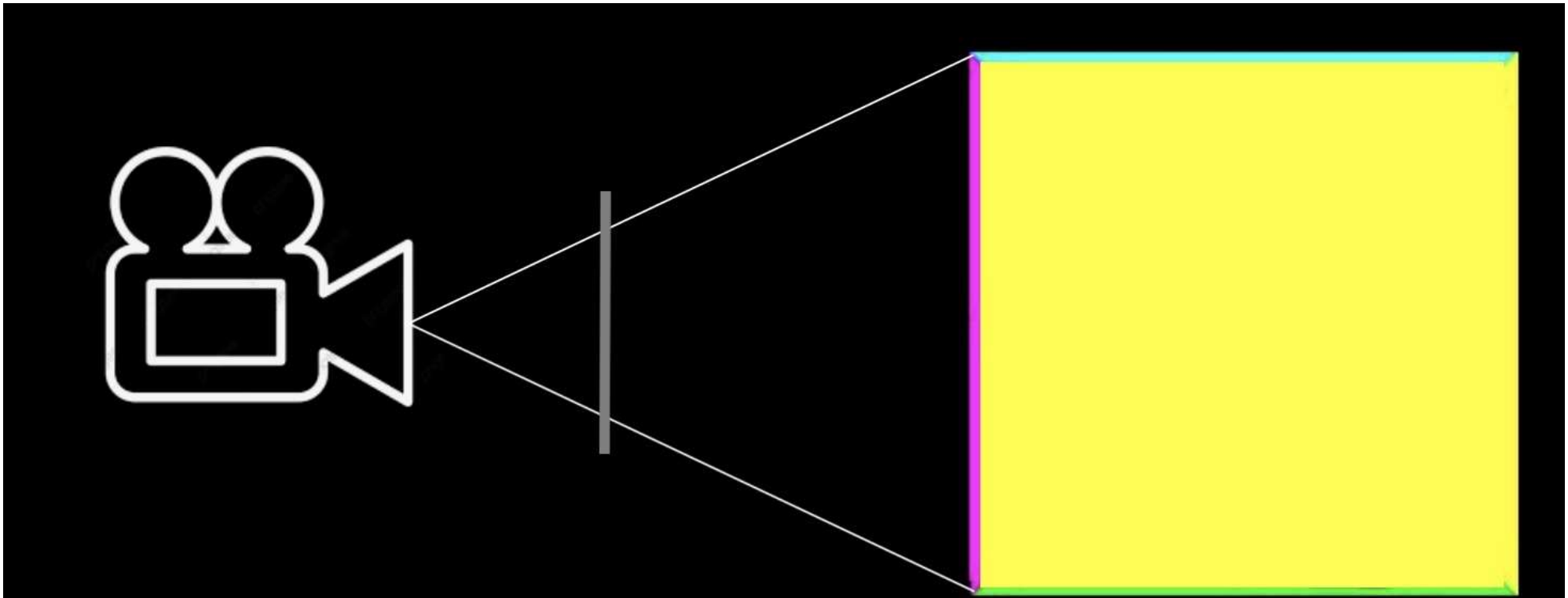
Focal Length of 1





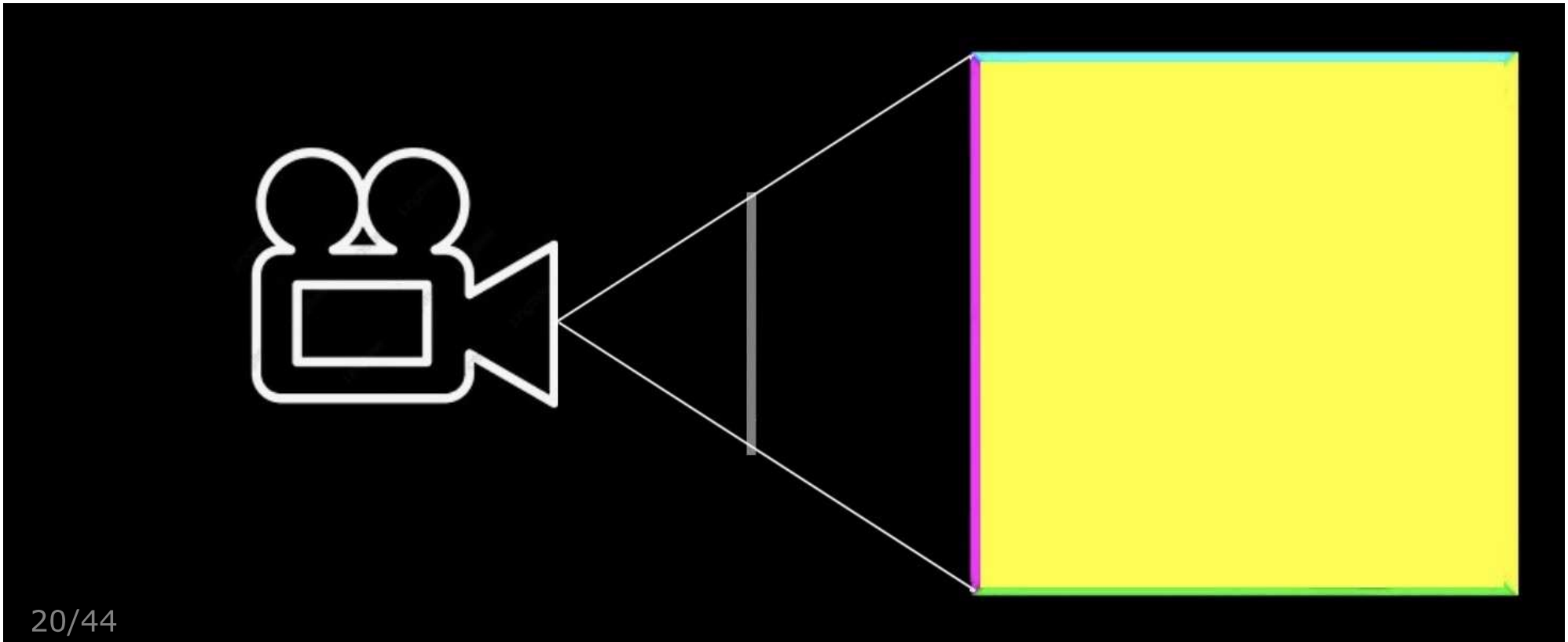
Now what if we move the POSITION of the camera ?

(without changing the focal length)

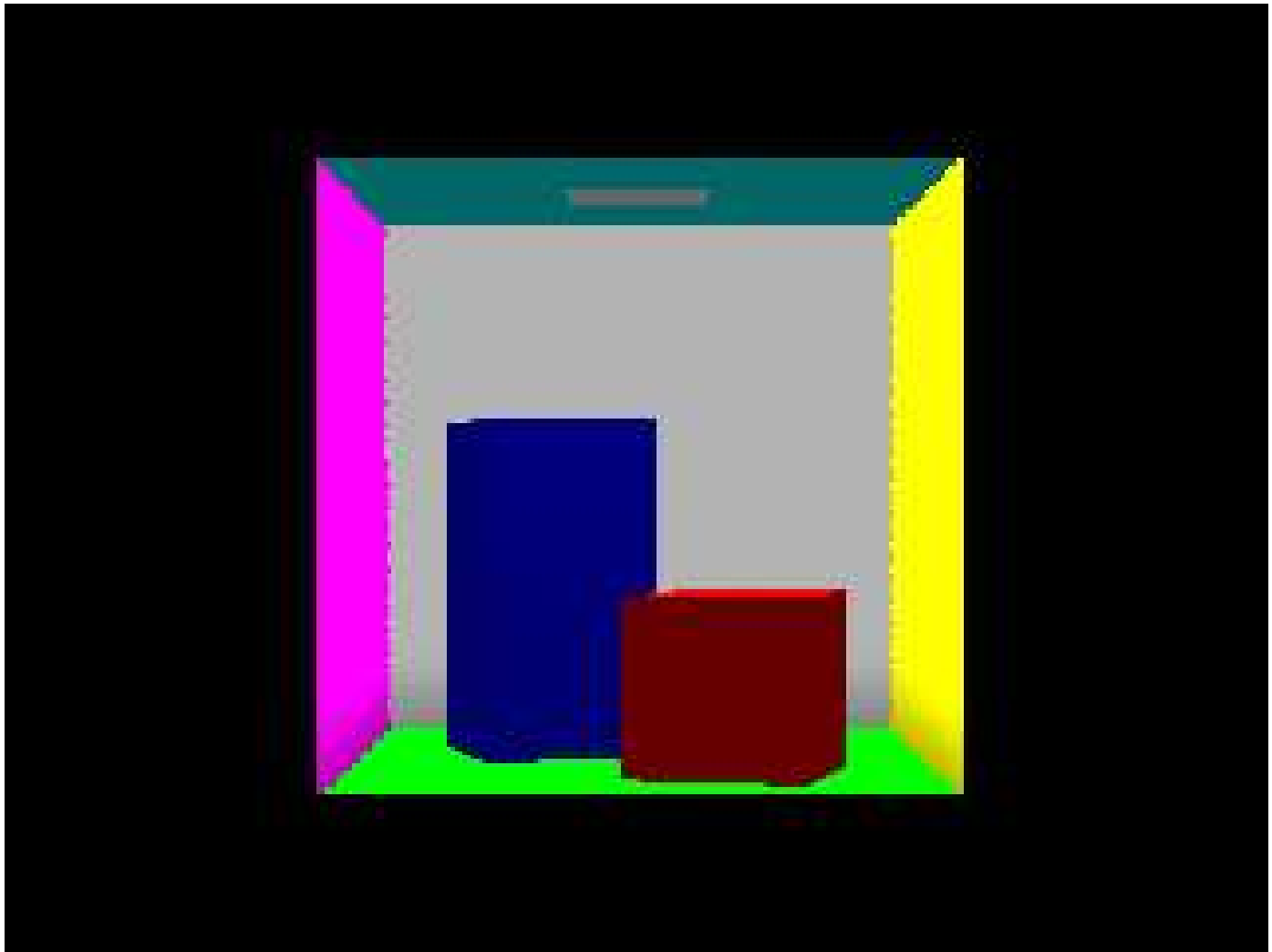


Again, the size of image on image plane will change

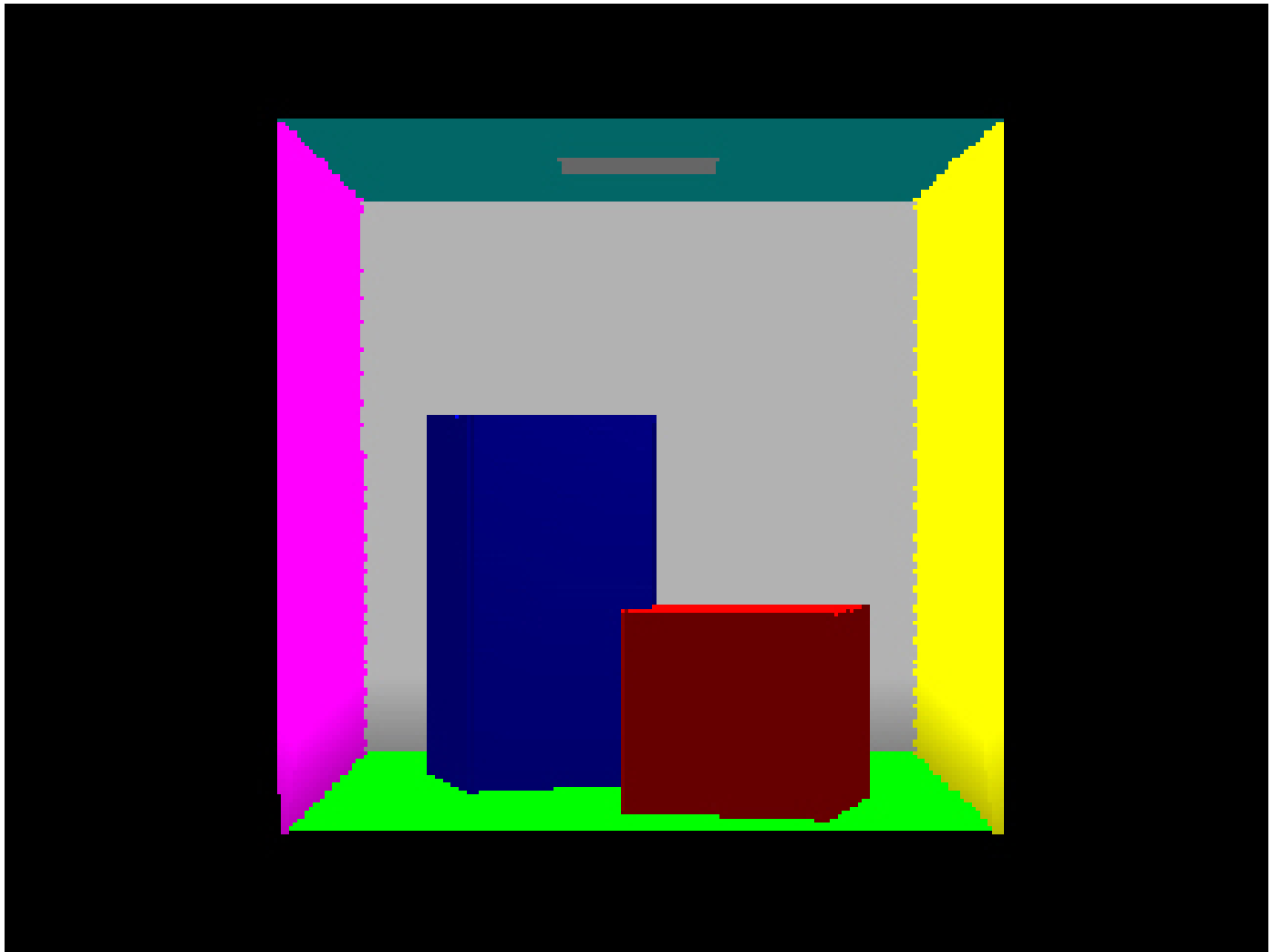
But ALSO so will our *perspective* on the scene !



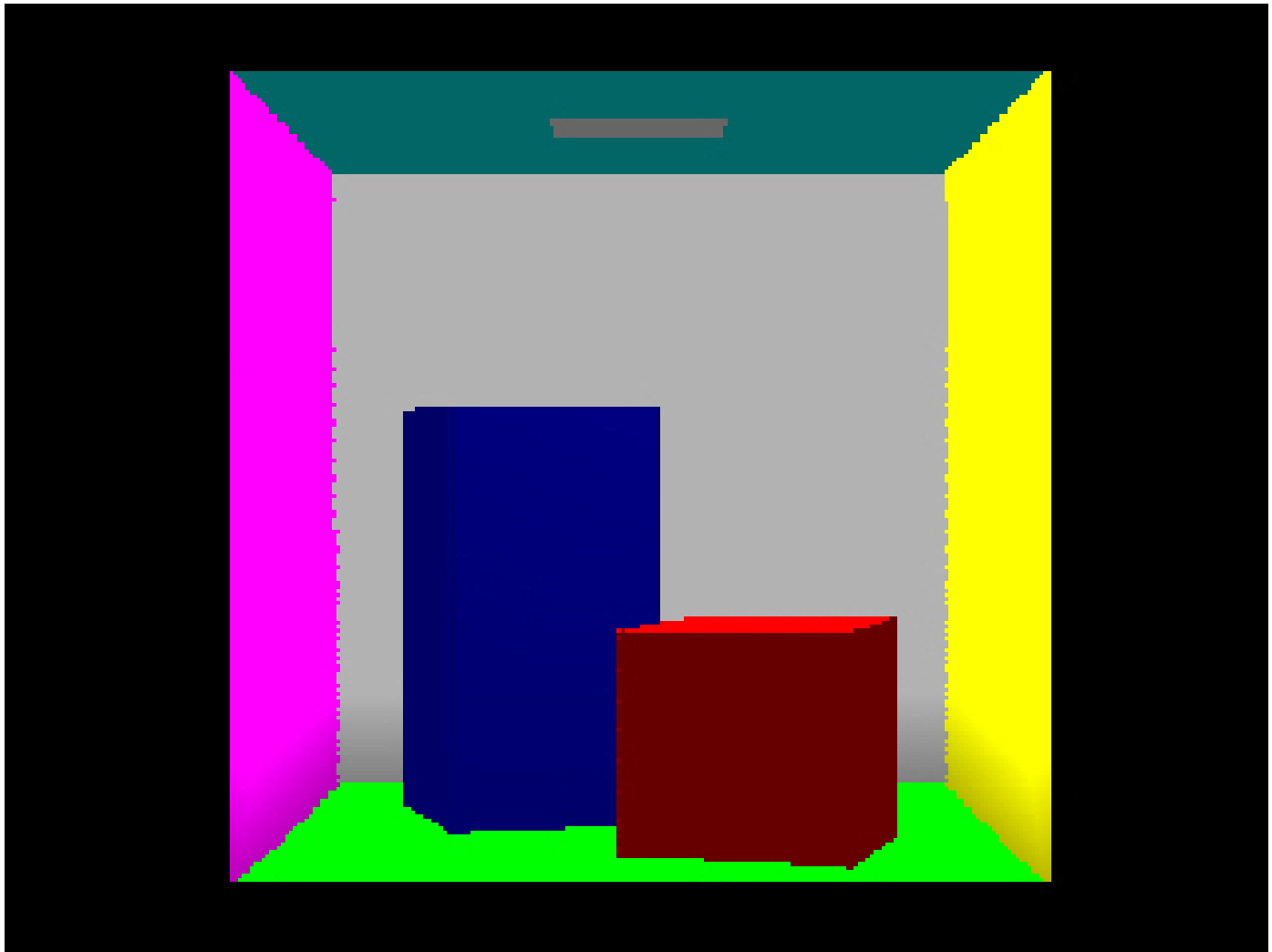
Camera Position #1



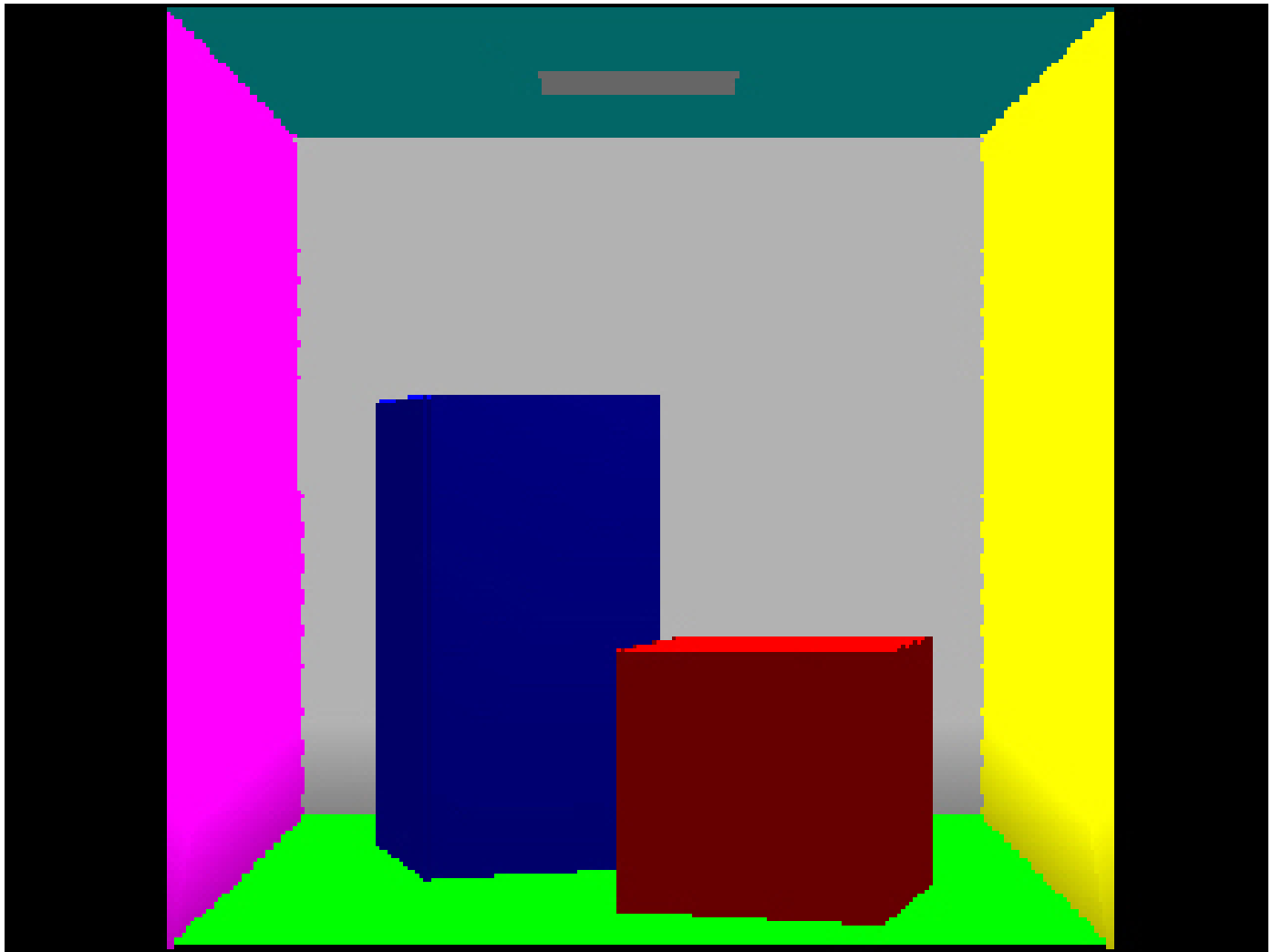
Camera Position #2



Camera Position #3



Camera Position #4



The change in perspective is hard to see

(because the size of the render is changing)

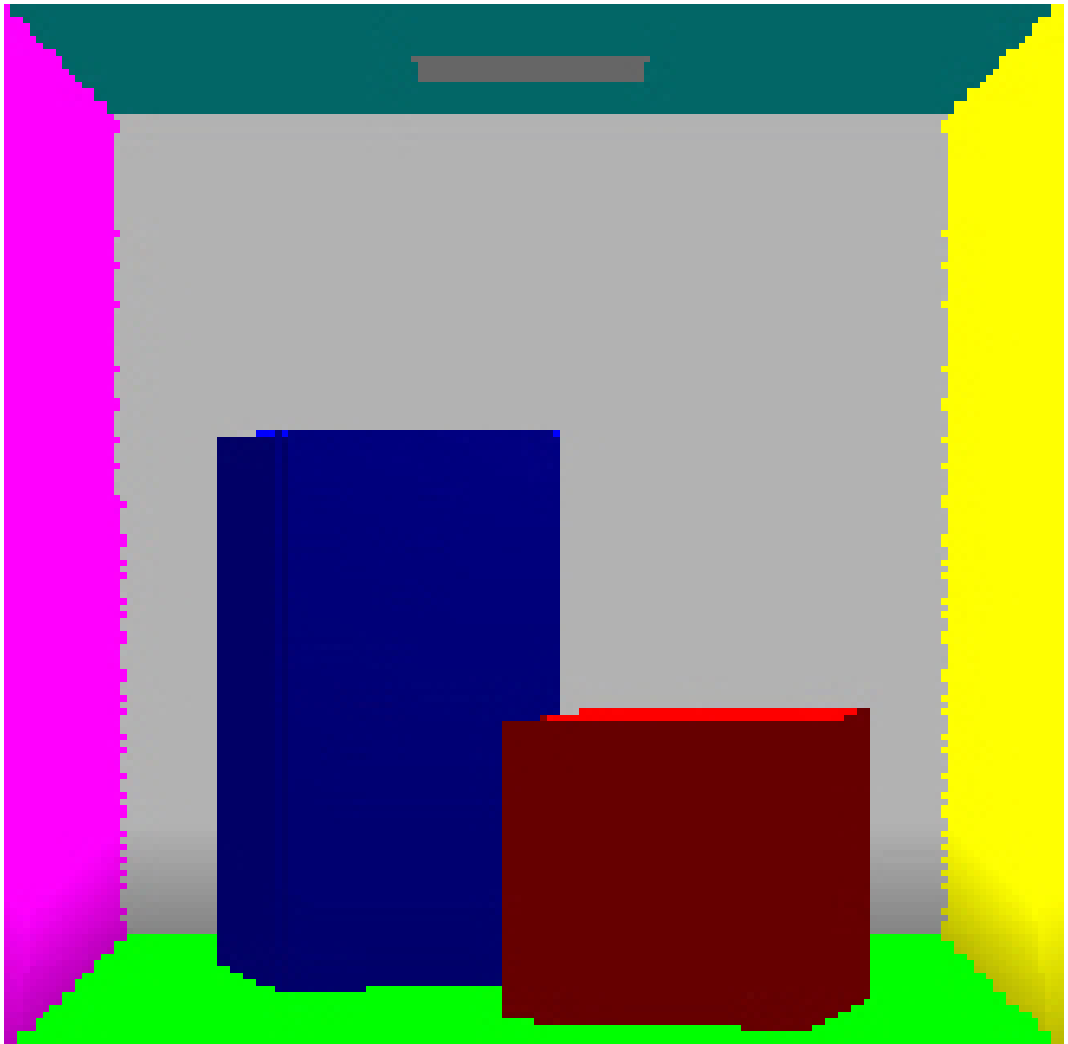
however...

If we perform BOTH operations at the same time
(Move camera position AND shift the image plane)

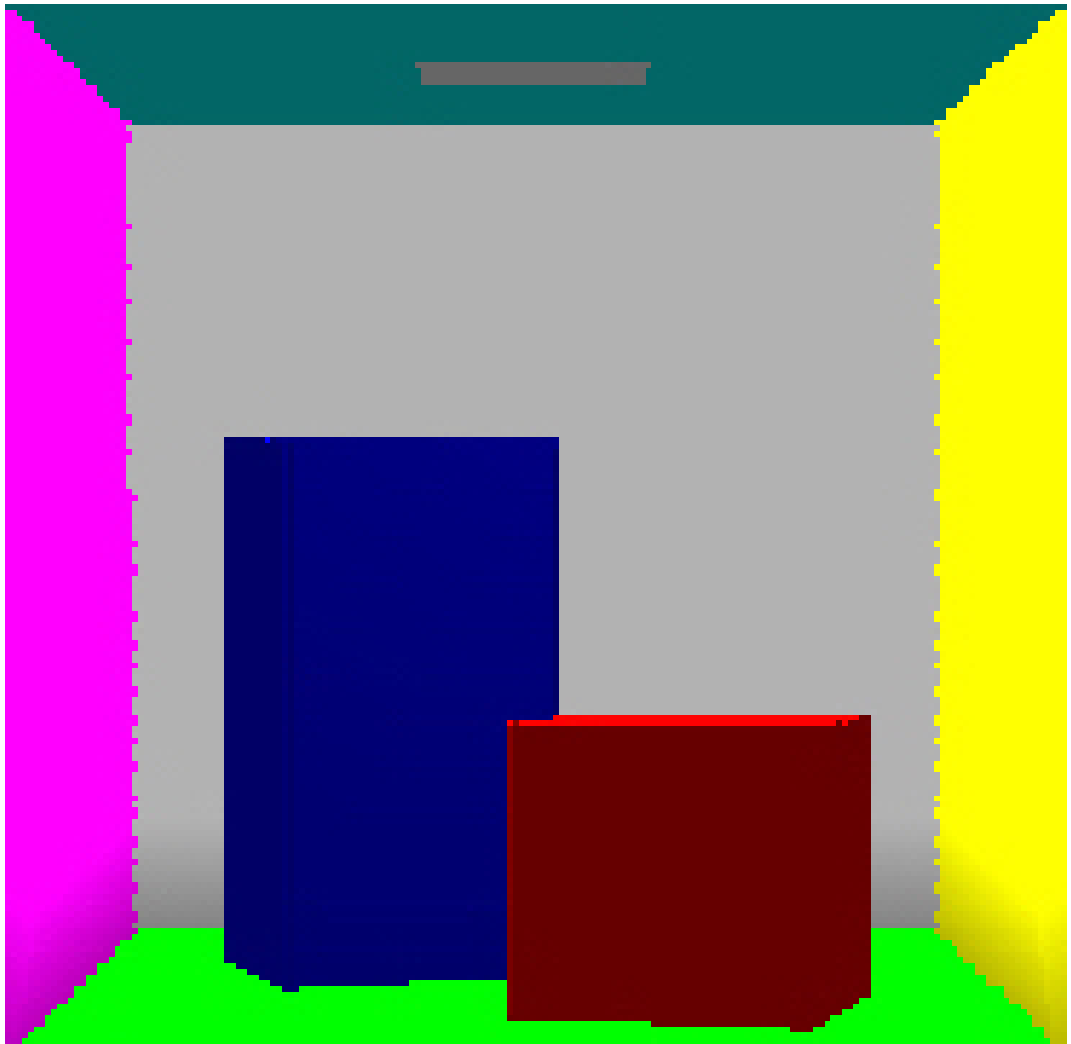
We can CANCEL OUT the two zoom effects
(providing we move/shift by appropriate distances !)

This will leave JUST the change in perspective...

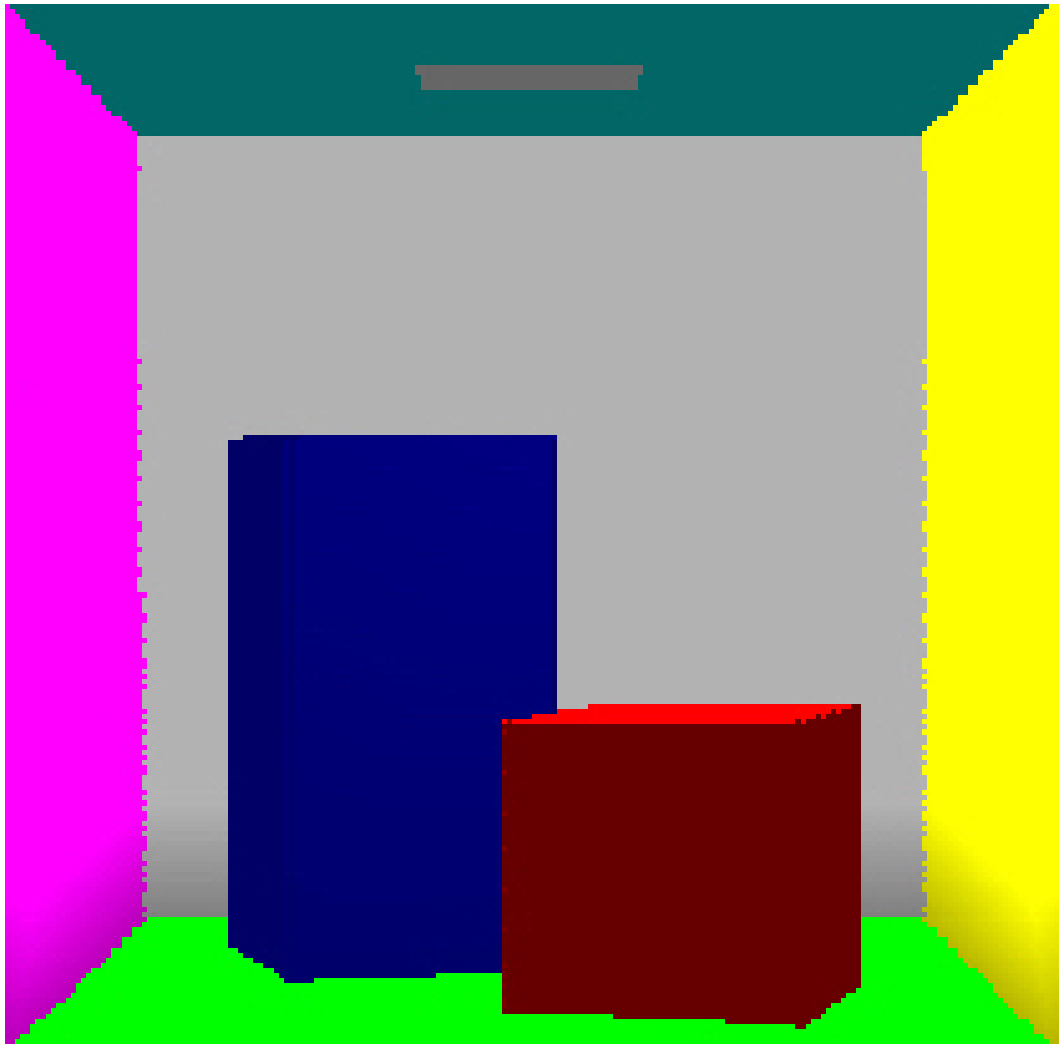
Change in perspective #1



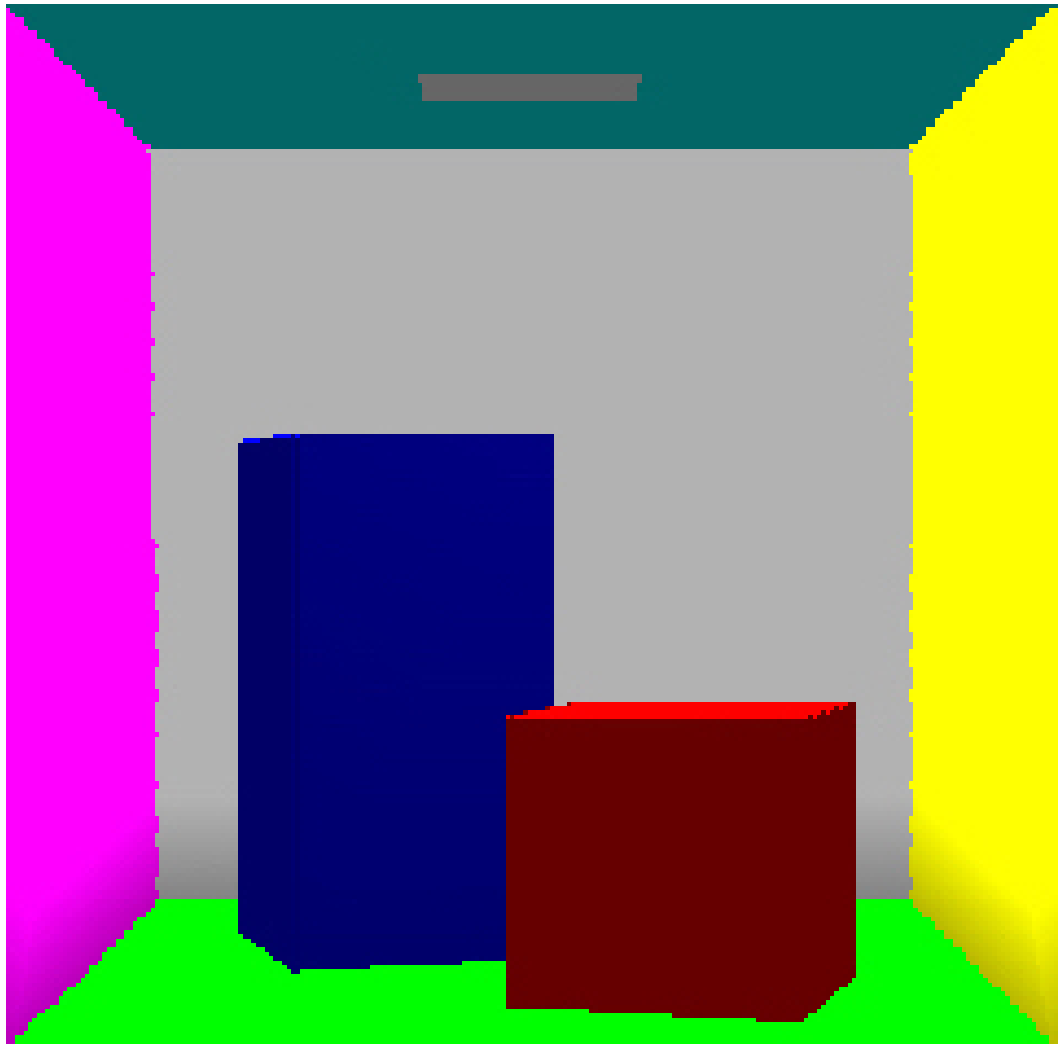
Change in perspective #2



Change in perspective #3



Change in perspective #4



Strangely familiar ?

Does this "deepening" effect look familiar ?
This effect is called a 'Dolly Zoom'

You will have seen it many times in film & cinema
For example in the 1970s horror classic:

jaws

https://www.youtube.com/watch?v=NB4bikrNzMk

Dolly Zoom #1



Dolly Zoom #2



Dolly Zoom #3



Dolly Zoom #4





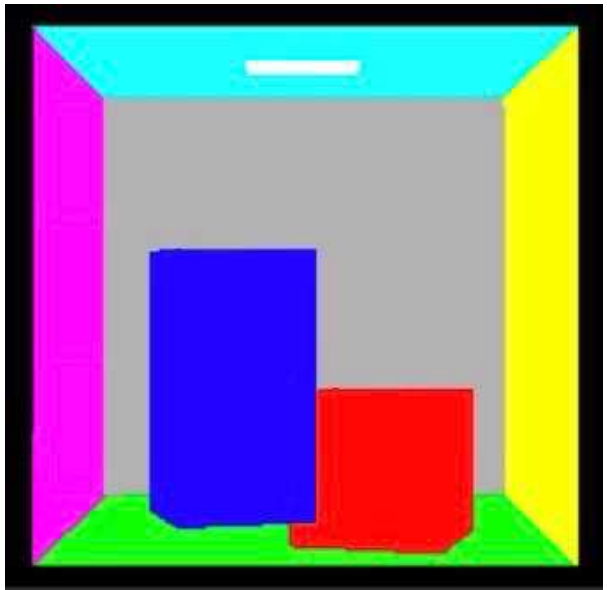
Problem with Z depth

When we 'down-project' model onto 2D image plane

We "lose" z depth of the vertices (and thus surfaces)

We must "reconstruct" z depth for correct layering

Workbook explains how to achieve this, to avoid:

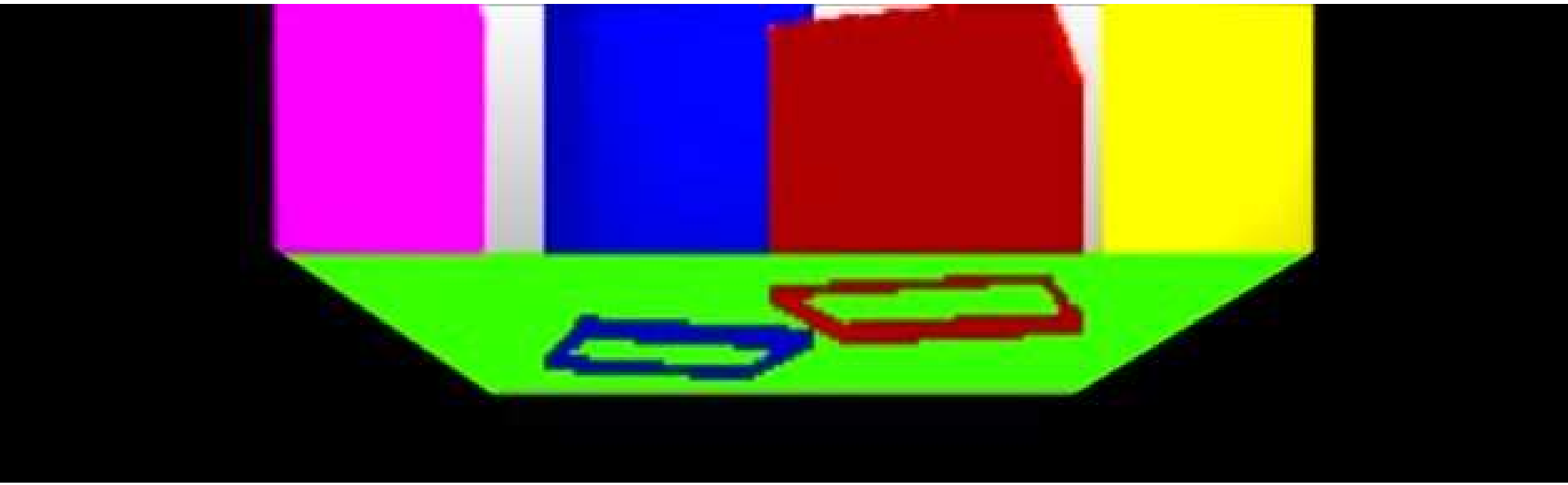


Oh noz ! My floor is broken !!!

It is normal to see an occasional "peek through"

(Due to close proximity and accumulated error)

However it shouldn't be as severe as the following...

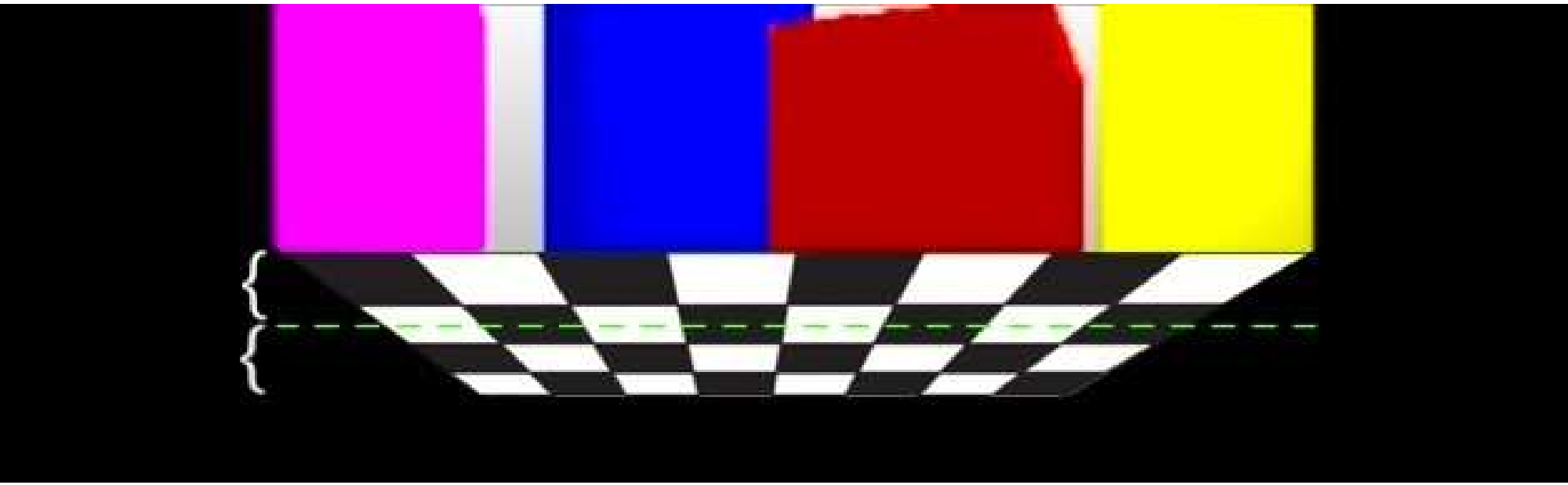


The reason ? Perspective !

Due to perspective, calculation of Z Depth is NOT linear

For given change in Y, change in Z is not consistent !!!

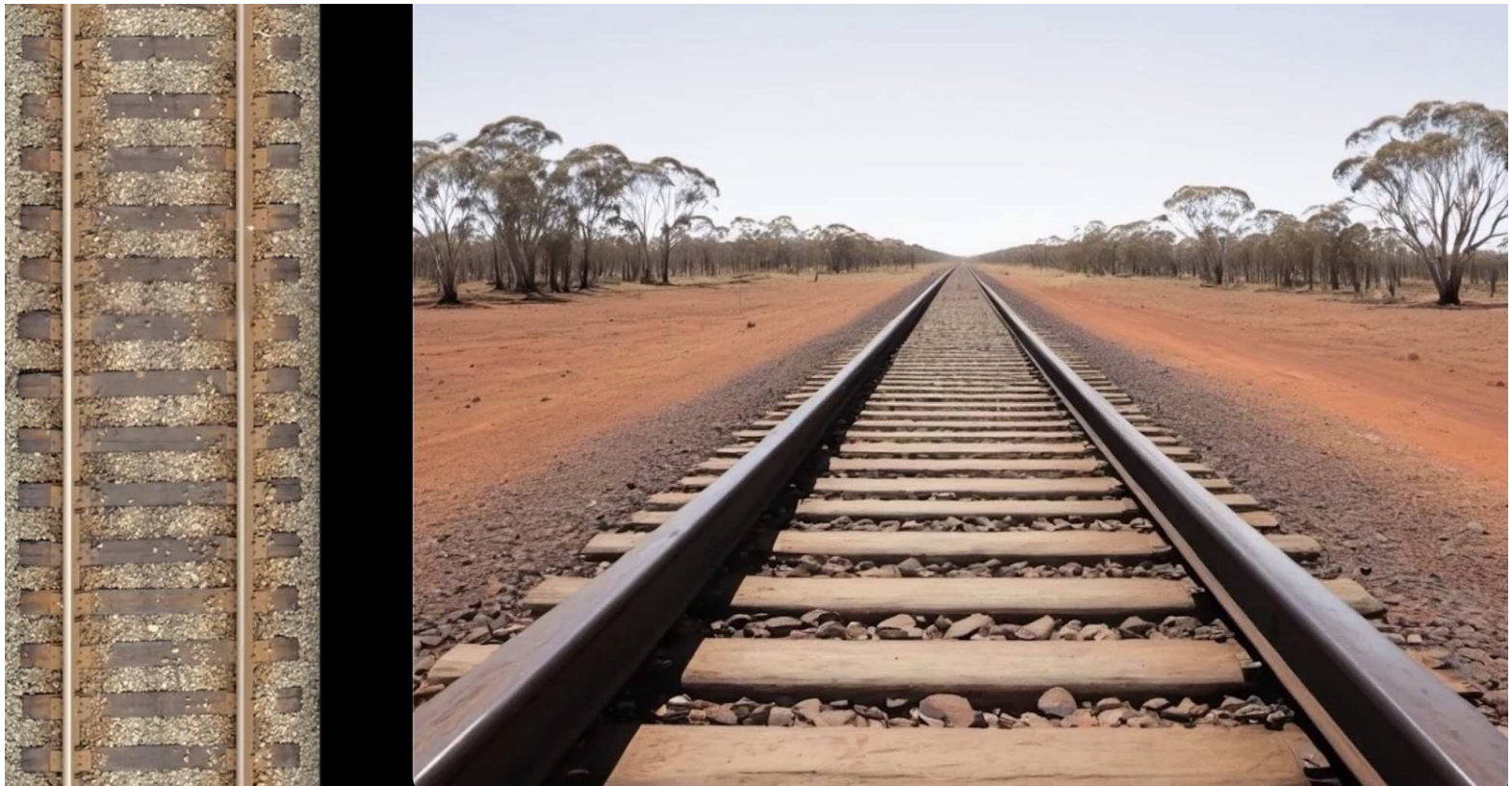
Dotted line is at half Y height of floor, but NOT half Z



An illustrative example

Consider a railway track: each 'sleeper' is the same size

But APPEAR as differing sizes, depending on Y position

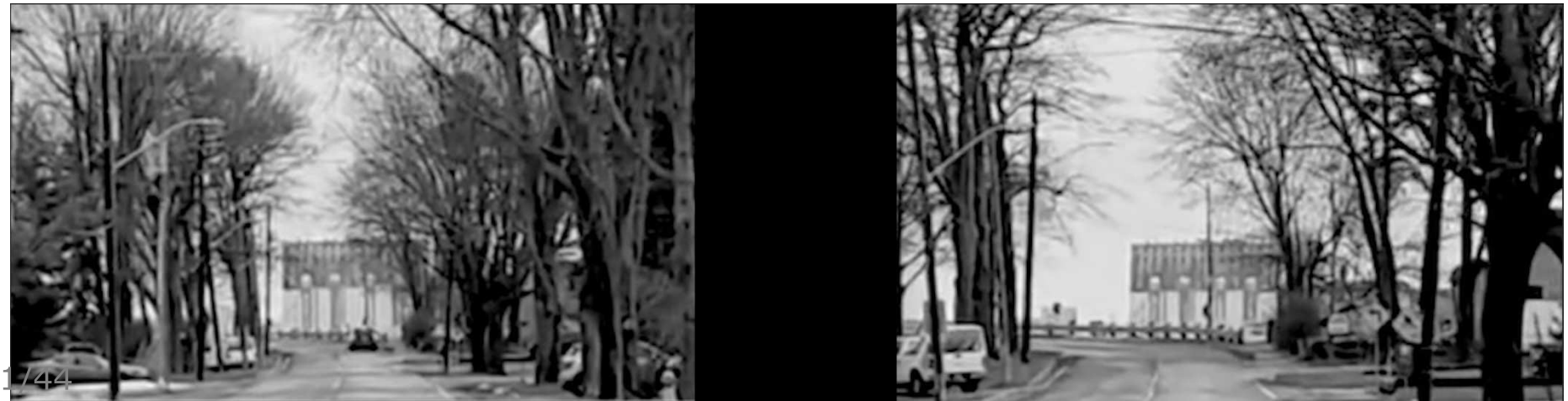


Perspective: more complicated than you think

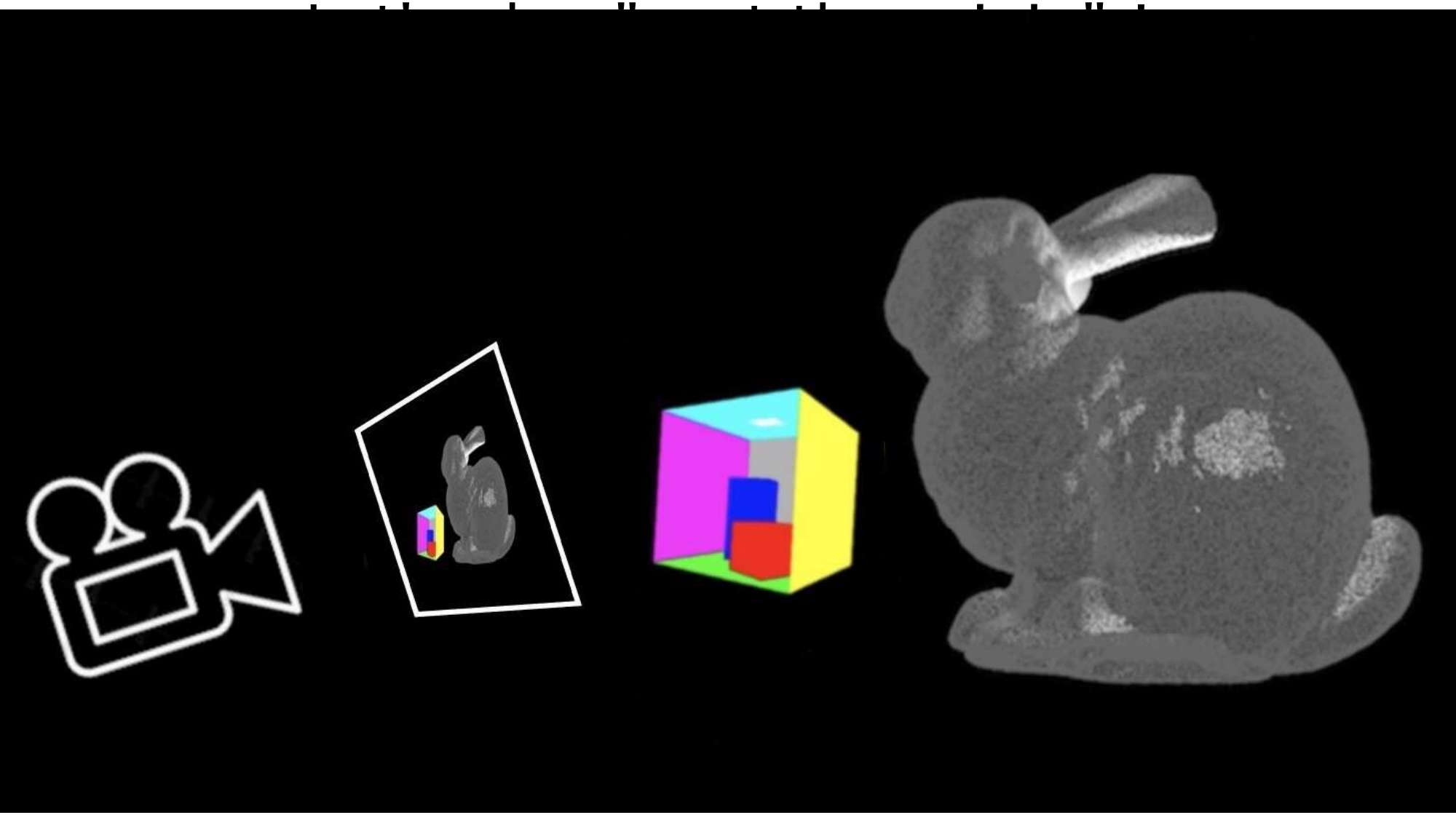
TheIncredibleShrinkingBuilding

Your homework task is to think about:

What causes the building to *appear* to shrink ?



Time for a quick game ?



Transposition

For many rendering features we will explore...

We must shift positions between coordinate systems

Locate a certain point RELATIVE to a specific origin

Helpful trivial example: Where is B relative to A ?

